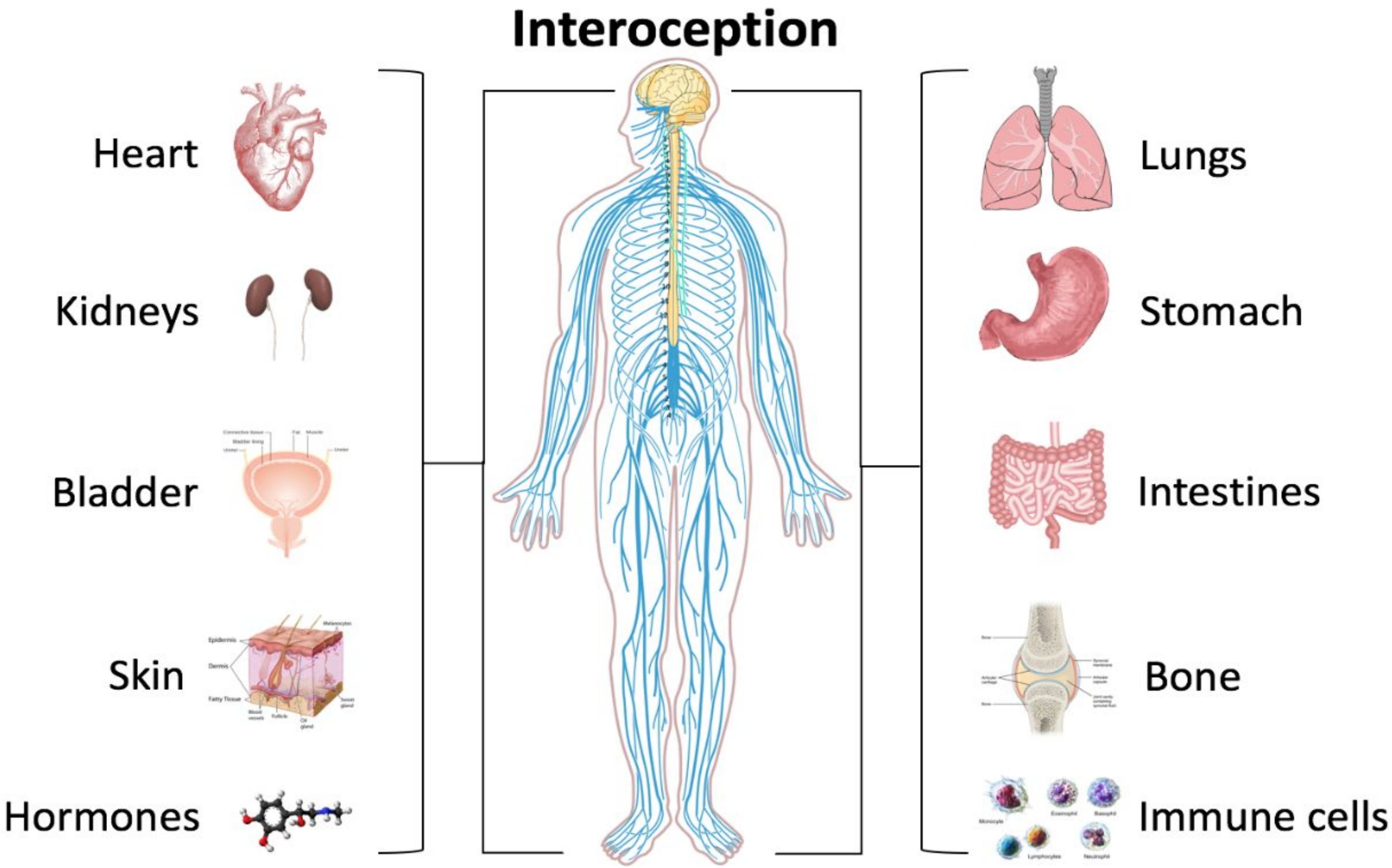
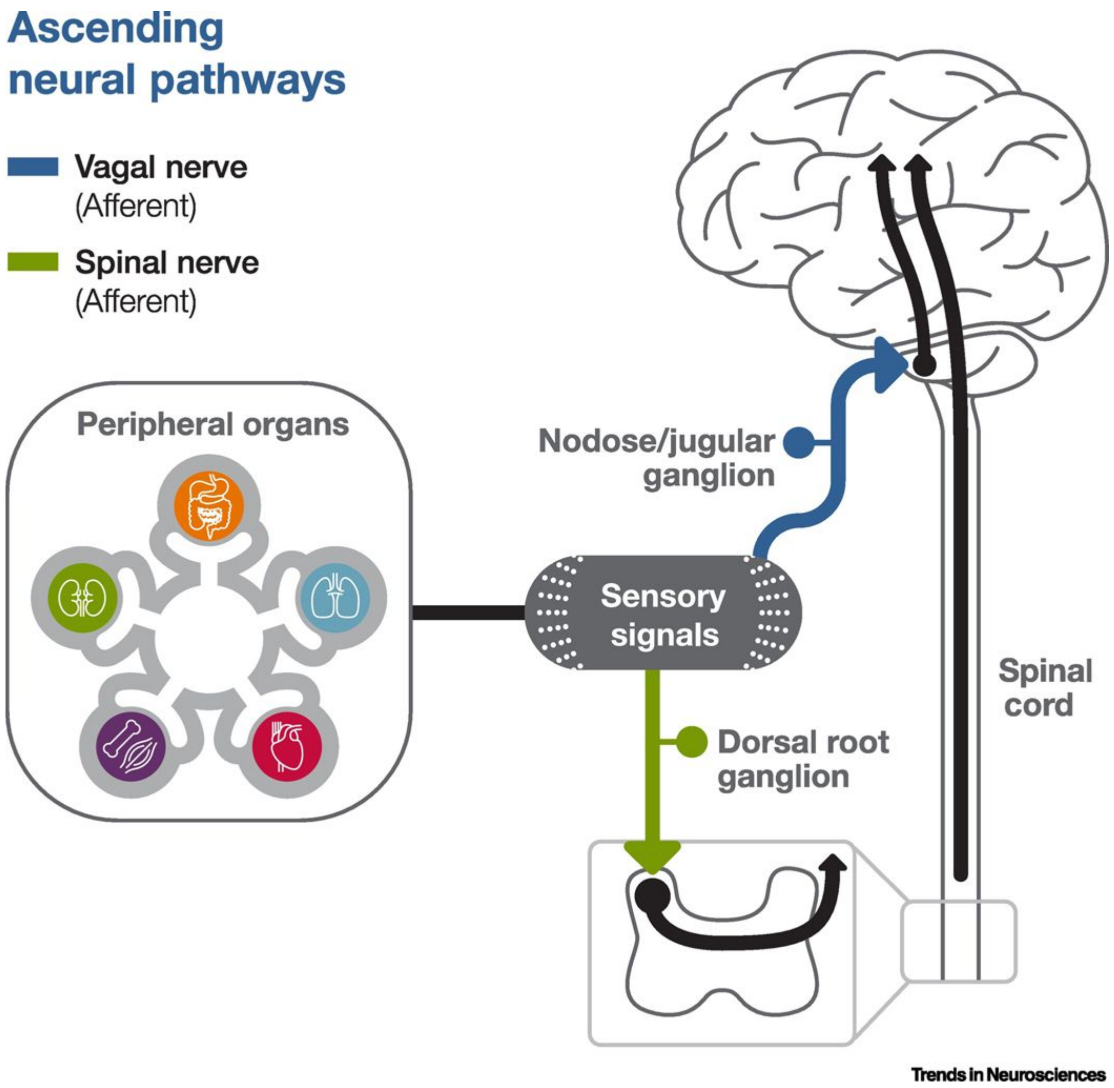
Modulating Autonomic Pathways

via Synthetic Acoustic and Haptic Stimulation

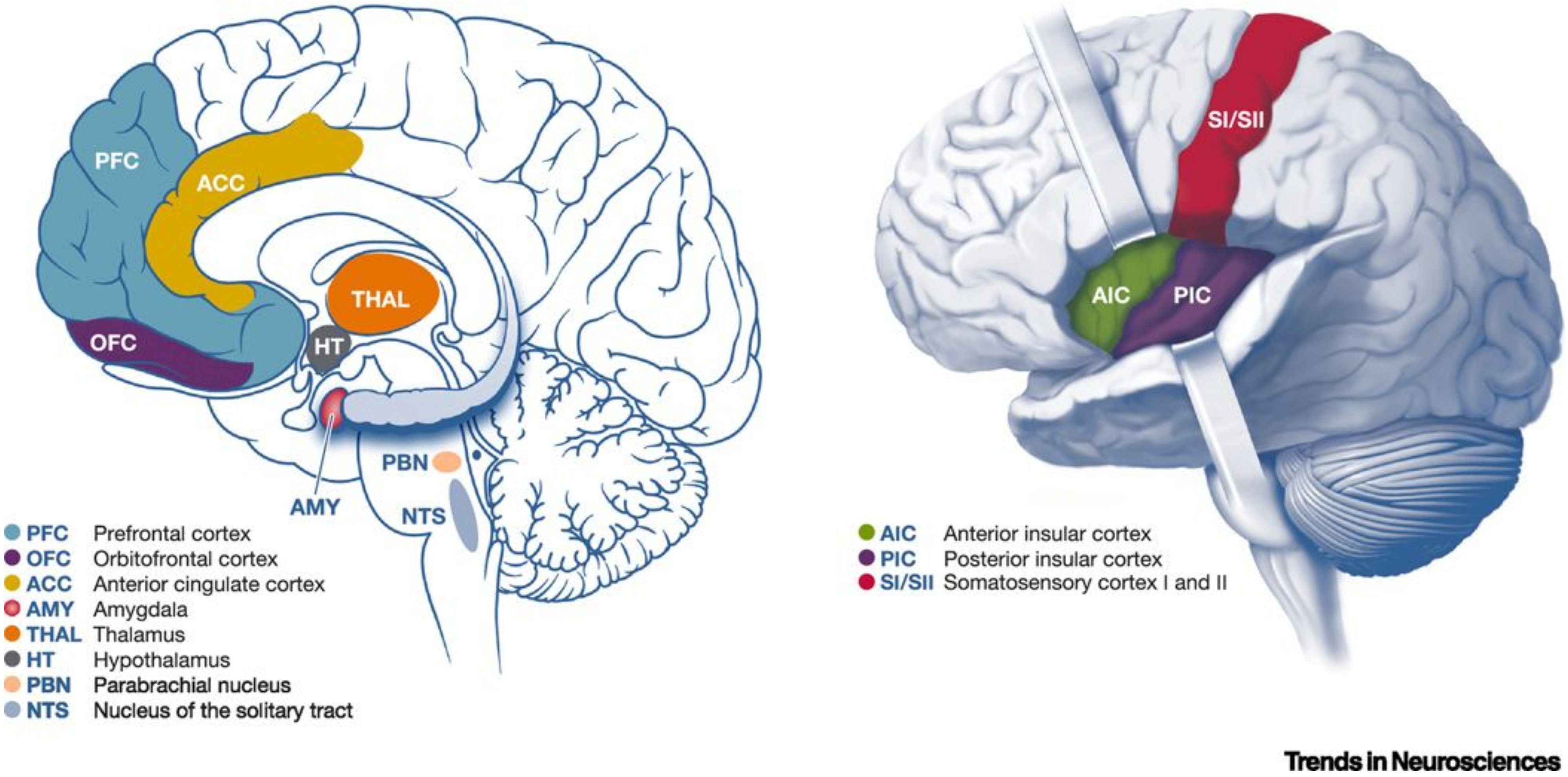
Shirin Ahmadi
Allostasis.io


Interoception
Heart
Kidneys
Bladder
Skin
Hormones
Lungs
Stomach
Intestines
Bone
Immune cells

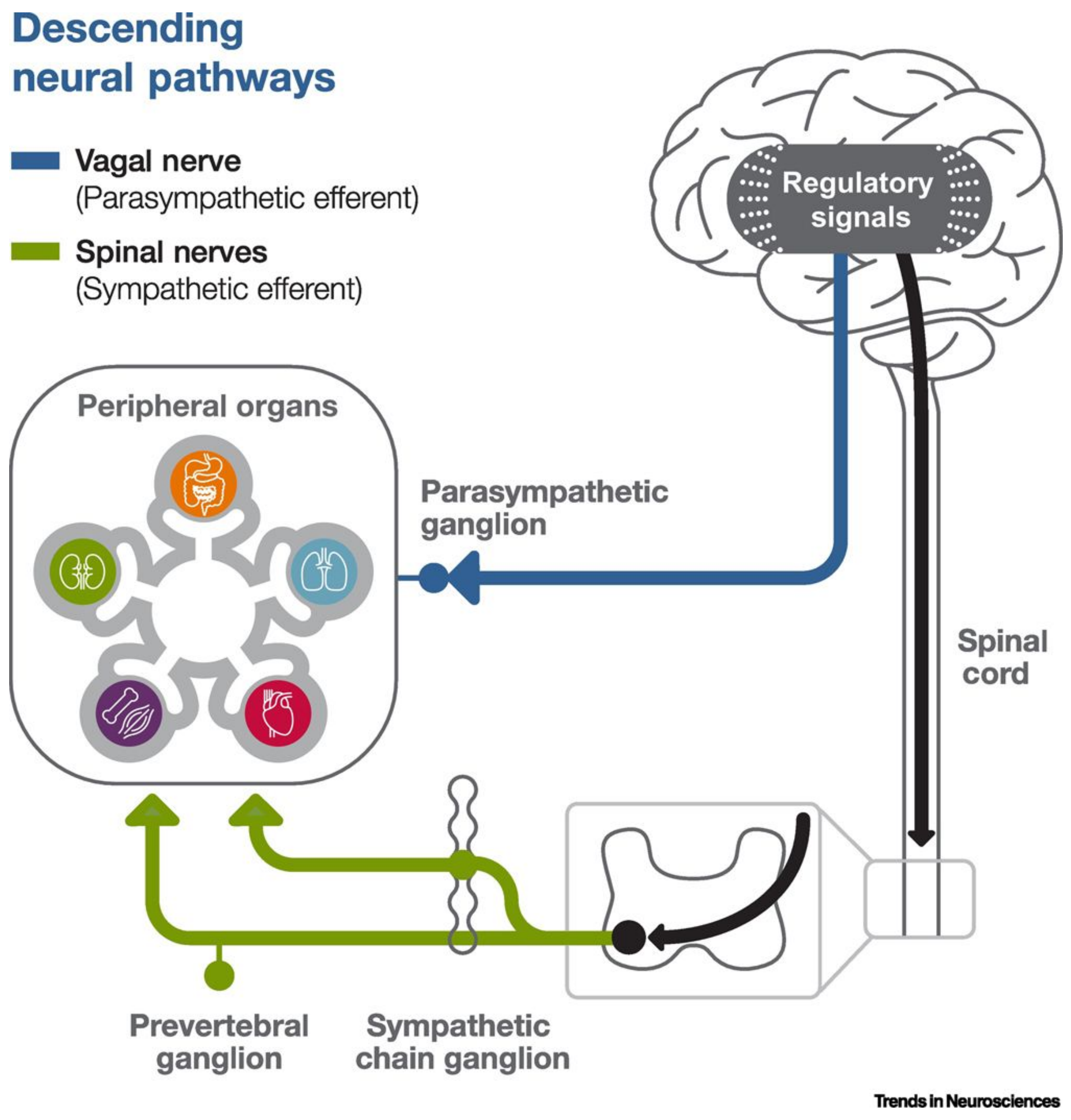
Illustrative diagram of sample ascending neural pathways of interoception



Brain regions involved in central processing of interoceptive signals.



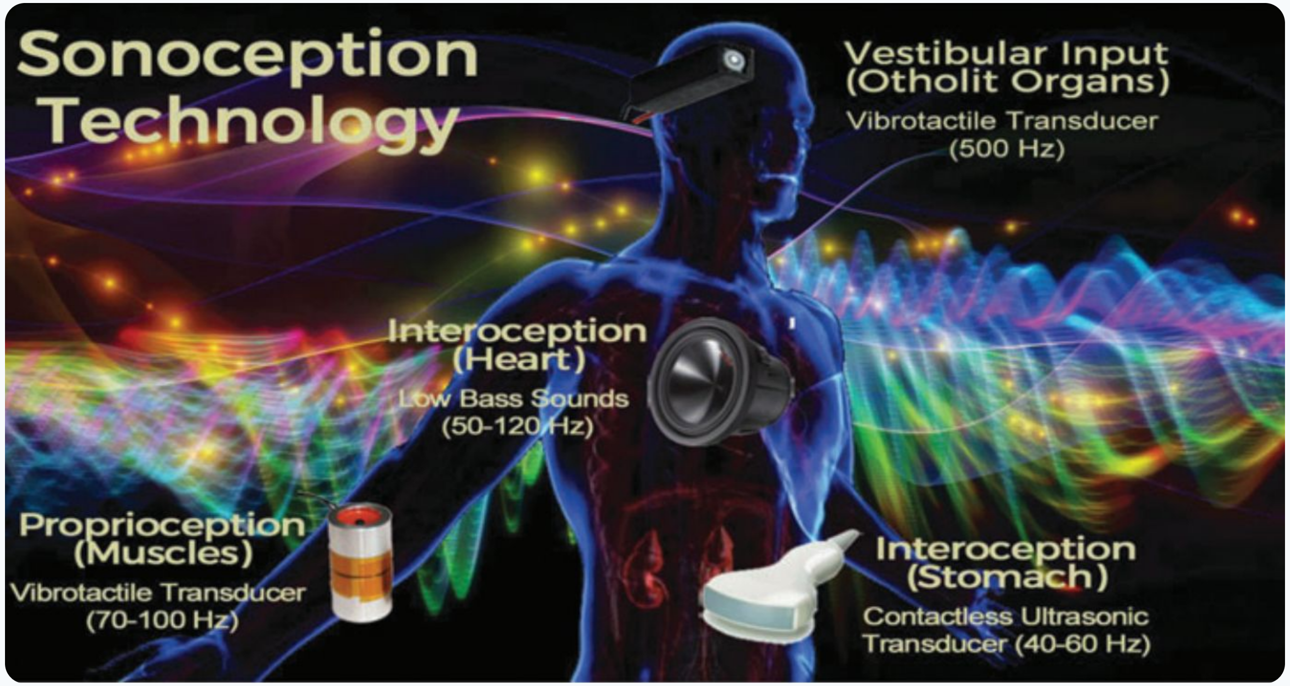
descending neural pathways of interoception



Interoceptive stimulation

Sonoception: Bridging the Gap in VR

Synthetic auditory stimulation of the insula alters interoceptive processing



Source: https://doi.org/10.31234/osf.io

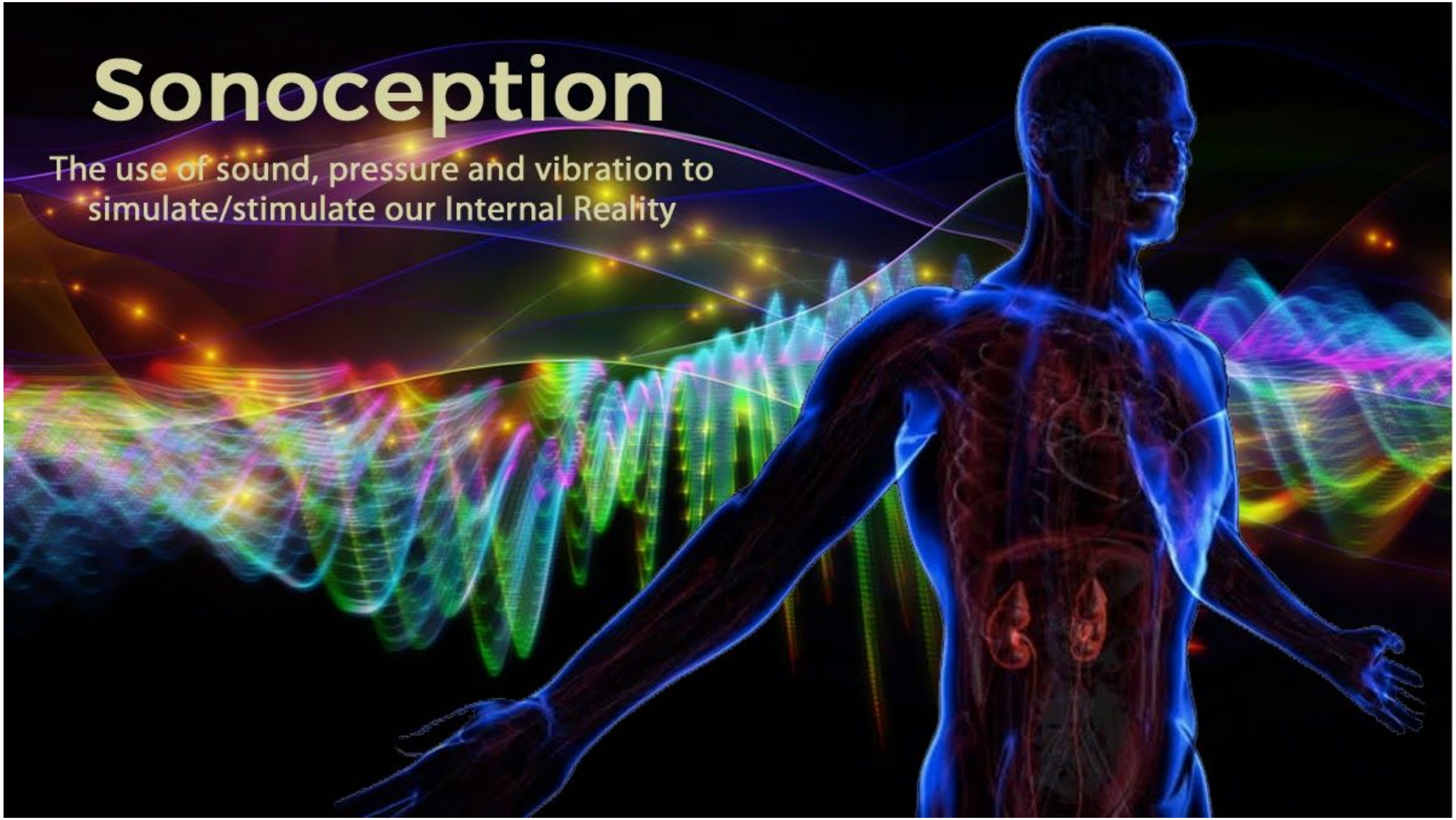
The "Sonoception" Paradigm

Current VR effectively reproduces **exteroceptive** (vision, hearing) and **proprioceptive** (motor) features, but lacks **interoceptive**/vestibular depth.

Sonoception uses wearable acoustic and vibrotactile transducers to:

- **Structure and Augment:** Modulate internal body awareness.
- **Targeted Stimulation:** Influence mechanoreceptors in the stomach, heart, and muscles.
- **Vestibular Integration:** Stimulate otolith organs for internal balance.

Sonoception acts as a noninvasive technological approach to replace or structure the contents of the inner body, enabling a fully immersive "Internal Reality."

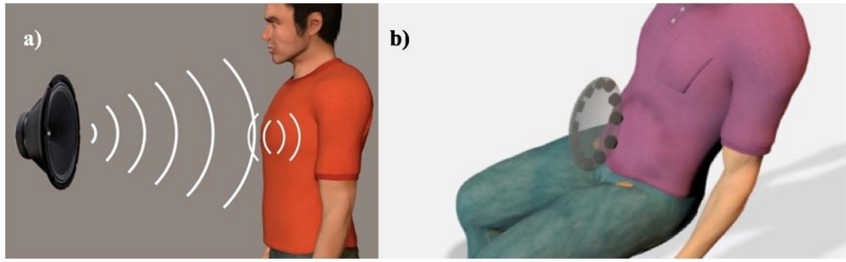

Sonoception
The use of sound, pressure and vibration to simulate/stimulate our Internal Reality

Interoception Module (Sonoception)

Stimulating internal body awareness through targeted acoustic technology.

Interoception (Stomach)

Optimize an ultrasonic transducer array (working from **40 to 60 kHz**) to employ acoustic levitation. This translates food particles, causing motion in the stomach walls to perceive internal movements.



Interoception Module (Sonoception). Employing contactless acoustic transducers; this novel technological paradigm will be able to modulate the interoception through the perception of movement in specific parts of the body such as the chest (a) and abdomen (b).

Interoception (Heart)

Low bass frequencies (between **50 to 120 Hz**) stimulate special receptors in the chest, allowing a user to distinctly feel their own heartbeat.

Employing contactless acoustic transducers, this technological paradigm enables the modulation of interoception in specific body parts such as the chest and abdomen.

The Insular Cortex & Interoceptive Pathways

Understanding the neural architecture of internal body awareness.

Neural Architecture

The **anterior insular cortex** receives information via a vast network of **C-fibers** connected to the Lamina I spinothalamocortical pathway.

This poly-modal system monitors:

- Vital States: Hunger, thirst, temperature
- Physical Inputs: Pain, itch, muscle contraction
- Systemic Activity: Hormonal, immune, and cardiorespiratory functions

C-tactile (CT) Perception

Parasympathetic Interoceptive Inputs

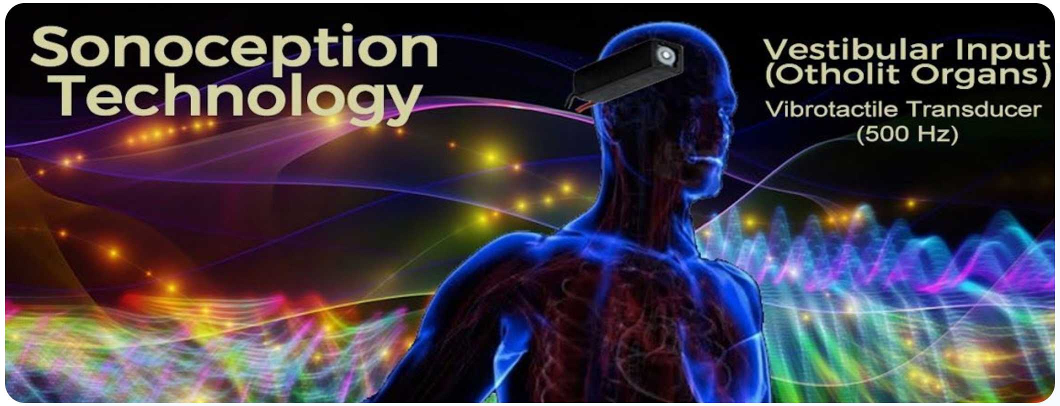
Recently discovered as a promising research field, **affective touch (CT)** is arguably the most significant stimulus.

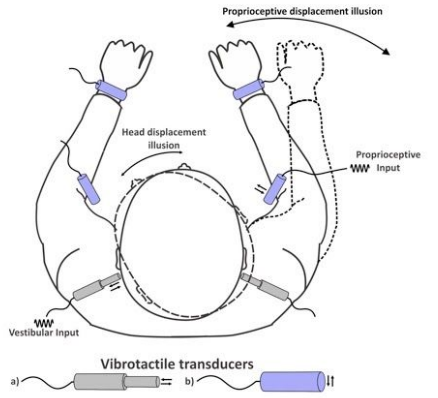
"CT afferent fibers constitute a secondary touch system with deep involvement in psycho-physiological pathways."

Source: Neural Mechanisms of Interoception and Affective Touch.

Proprioception & Vibrotactile Input

Modulating body awareness through cutaneous and tendon stimulation.





Cutaneous Receptors

Located in the skin around finger, elbow, ankle, and knee joints, these receptors provide both exteroceptive and proprioceptive information. Similar to muscle spindles, they encode movement kinematics and show directional sensitivity.

(Lee et al, 2013)

Vibrotactile Tendon Stimulation

Applying 70–100 Hz vibrations to biceps/triceps tendons of immobile limbs generates illusory sensations of arm displacement. Increasing frequency directly correlates with higher perceived movement velocity.

(Naito et al. 1999; Roll and Vedel, 1982)

Sonoception: Employing vibrotactile transducers to modulate proprioception through muscle vibrations and the vestibular input through bone-conduced vibrations.

Sonoception demonstrates the potential to activate the insular cortex (**a brain region involved in interoceptive processing**) in a non-invasive manner and holds promise for advancing our understanding of the **interoceptive system.**

Sonoception: Synthetic auditory stimulation of the insula alters proxies of interoceptive processing at behavioral and cortical levels.

Daniele Di Lernia[1], Andrea Zaccaro[2,a], Aleksandra Herman[3,a], Valerio Villani[4], Gianluca Finotti[4], Marcello Costantini[5], Francesca Ferri[6], Manos Tsakiris[4,7,b], Giuseppe Riva[1,8,b]

[1] Humane Technology Lab, Università Cattolica del Sacro Cuore, Milan, Italy.
[2] Department of Psychological, Health and Territorial Sciences, "G. d'Annunzio" University of Chieti-Pescara, Chieti, Italy.
[3] Laboratory of Brain Imaging, Nencki Institute of Experimental Biology of the Polish Academy of Sciences, Warsaw, Poland.

Sonoception uses synthetic auditory stimulation to activate the insular cortex. The technique involves using headphones to deliver synthetic auditory frequencies at 6Hz and 2Hz to activate the left and right insula, respectively. The paper reports that Sonoception produces measurable neural and behavioral effects.

The paper concludes that Sonoception is a promising non-invasive technique to manipulate the interoceptive system and advance our understanding of its role in emotion, cognition, and psychopathology.

https://pmc.ncbi.nlm.nih.gov/articles/PMC5519604/

Altering the experience of being in a body, with the goal of improving the well-being of patients.

•**Embodied Medicine**: the use of technologies to alter the experience of being in a body, with the aim of improving health and well-being.

•**Locked-in Syndrome**: neurological condition called locked-in syndrome (LIS), in which patients are completely paralyzed except for eye movements, but retain consciousness and sensation.

Commentary: Embodied Medicine: Mens Sana in Corpore Virtuale Sano

Francesca Pistoia [1], Antonio Carolei [1], Simona Sacco [1] and Marco Sarà [2]*

[1] Department of Biotechnological and Applied Clinical Sciences, Neurological Institute, University of L'Aquila, L'Aquila, Italy,
[2] Post-Coma Intensive Rehabilitation Care Unit, San Raffaele Hospital, Cassino, Italy

Keywords: body matrix, locked-in syndrome, body representations, virtual reality, motor imagery

A commentary on

Embodied Medicine: Mens Sana in Corpore Virtuale Sano
by Riva, G., Serino, S., Di Lernia, D., Pavone, E. F., and Dakanalis, A. (2017). Front. Hum. Neurosci. 11:120. doi: 10.3389/fnhum.2017.00120

We read with interest the recently published paper about the potential role of "*embodied medicine*" (Riva et al., 2017). Authors suggest the use of advanced technologies for altering the experience of being in a body, with the goal of improving the well-being of patients. This paradigm is intriguingly summarized through the key message "*Mens Sana in Corpore Virtuale Sano*" and is recommended for patients with different neurological and psychiatric disorders including neglect,

•**Body Representation Disorder**: The paper argues that LIS patients suffer from a disorder of body representation, due to the **lack of efferent (outgoing) signals from the brain to the body** and the cerebellum, which affects their cognition, emotion, and behavior.

•**Sonoception Technology**: which uses vibrotactile stimuli to create a sensation of body movement and to train the self-monitoring of patients.

https://pubmed.ncbi.nlm.nih.gov/33378657/

HHS Public Access

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Interventions and Manipulations of Interoception

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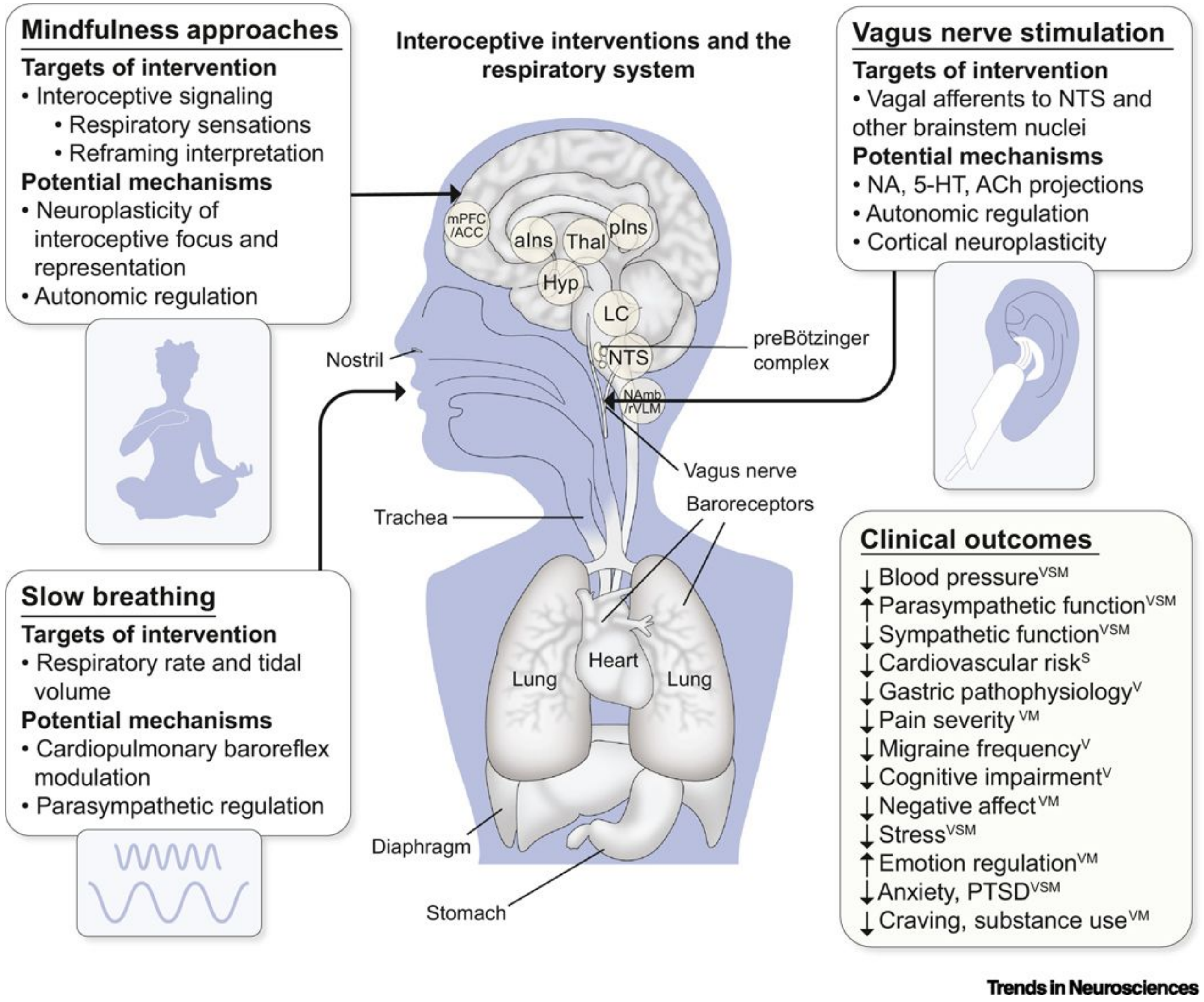


EMBODIED THEORIES OF EMOTION

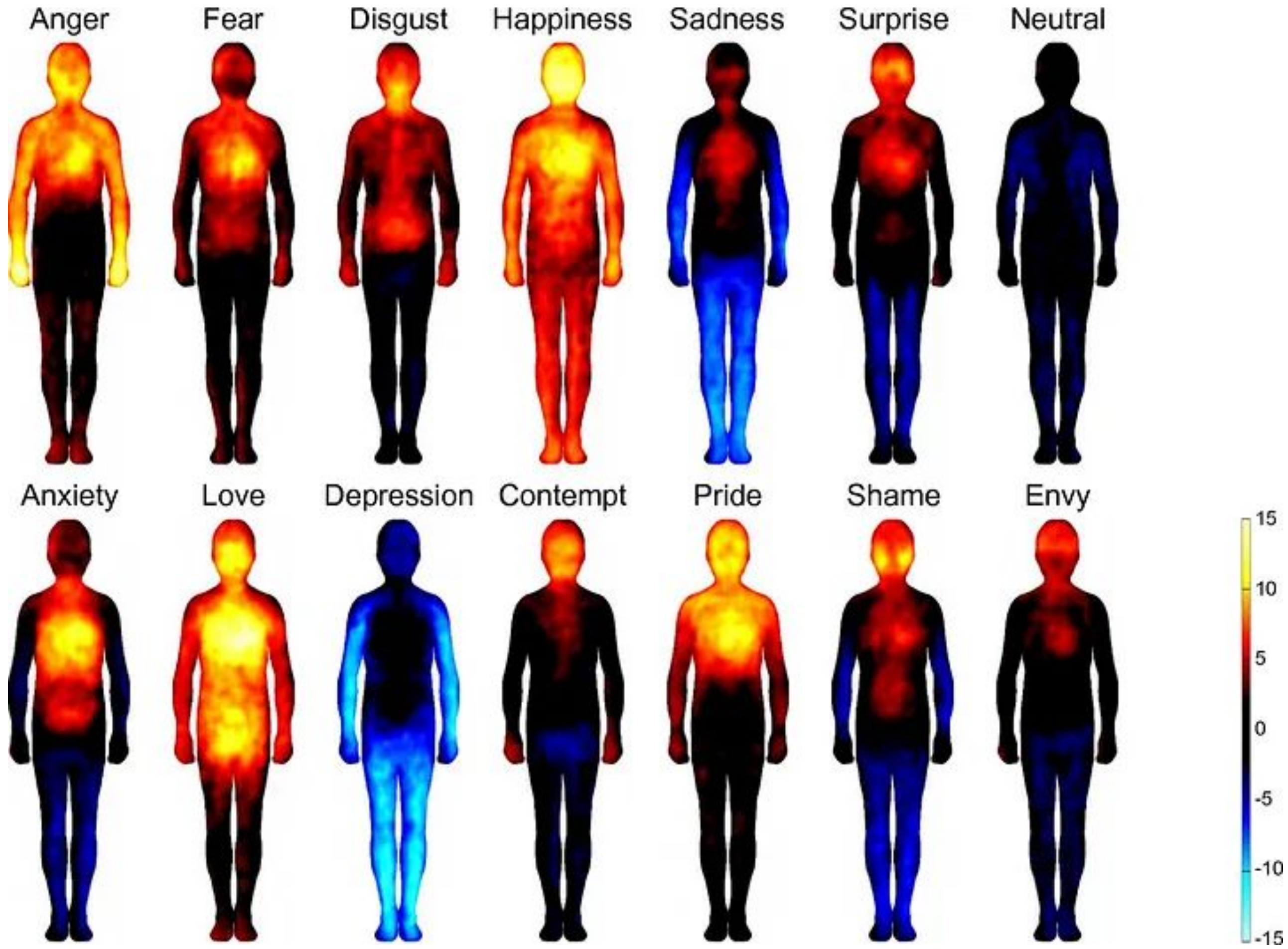
Emotions are **generated by the body** and **sensed by the autonomic nervous system** to create a conscious experience.

Core Investigation

Can emotional interoceptive feedback from a **virtual body** in VR be used for **emotional regulation**?

- Body Ownership Illusion
- Interoceptive Feedback Loops
- VR-based Regulation Techniques

[I felt it too: Emotion regulation through Augmented interoception. | by Vinit Mankar | Vinit Mankar | Medium]



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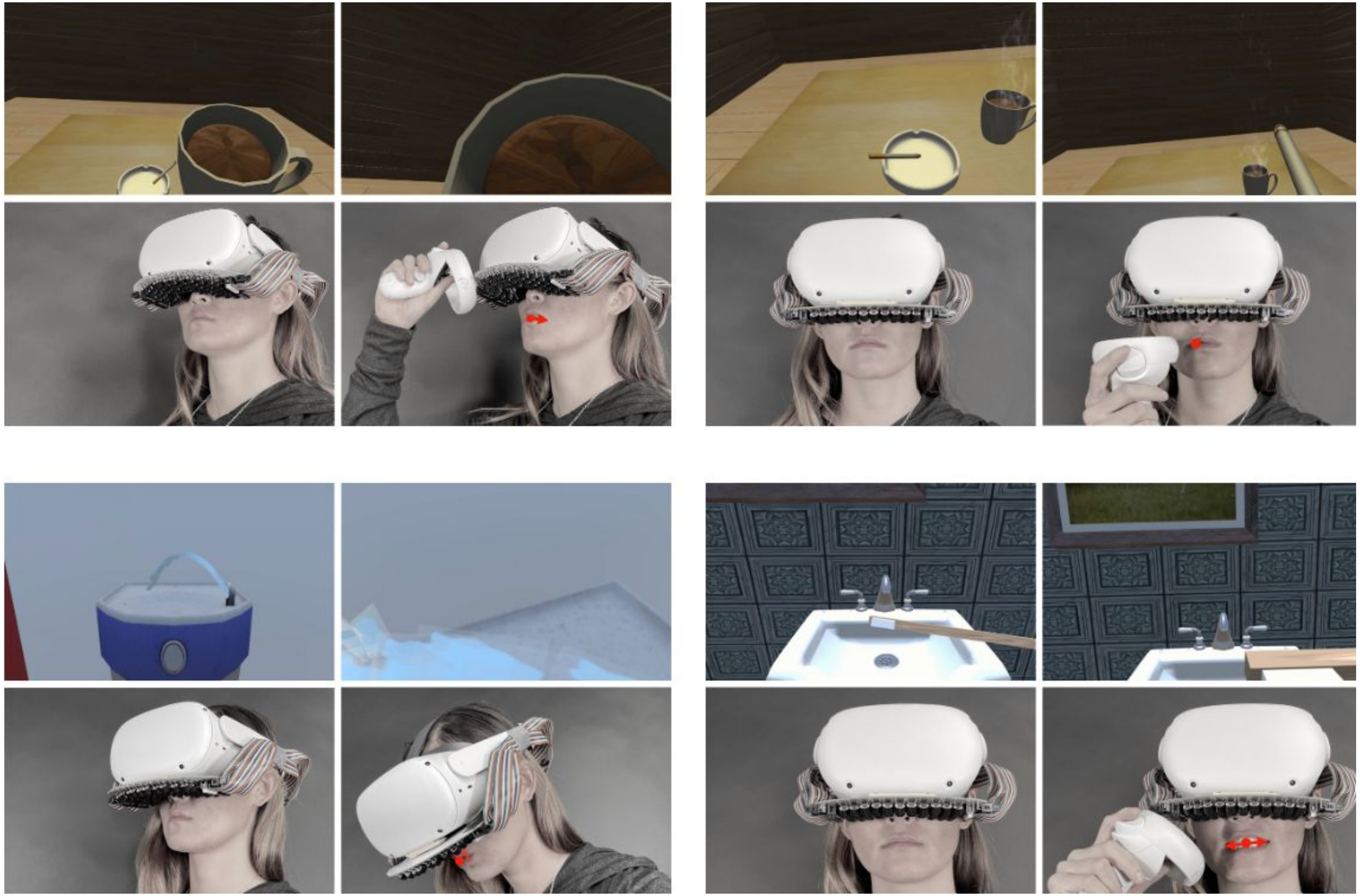

POOR INTEROCEPTIVE AWARENESS

Clinical Conditions & Disorders Associated with Low Interoceptive Insight:

- Anxiety
- Depression
- Sensory Processing Disorder (SPD)
- ADHD
- Eating Disorders
- Obesity
- Schizophrenia
- Dementia
- OCD
- Trauma Disorders
- Panic Disorder
- Suicide Attempters & Planners
- Drug & Alcohol Addiction
- Chronic Pain Syndromes
- Autism